Bilateral Trade with Lying Costs

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Abstract

Consider the bilateral trade setting in Myerson and Satterthwaite 1983, but the buyer and seller have a "lying cost" characterized by a disutility in the form of $\gamma_i|t-t'|$, where γ_i is agent i's lying cost parameter, t is the agent's true value, and t' is the agent's reported value. I provide some preliminary results on the implementability of the first-best, efficient allocation rule.

1 Introduction

In the bilateral trade setting described in Myerson and Satterthwaite 1983, it is impossible to implement the efficient allocation rule with an incentive compatible, individual rational, and budget balanced mechanism. This assumes that agents treat their report as a strategic decision concerning only their possible allocation and payment. I consider a setting where both the buyer and seller have commonly-known lying cost parameters, γ_B, γ_S respectively. With a sufficiently high total lying cost $\gamma_B + \gamma_S \geq 1$, fully efficient trade is implementable for any value distributions. With $\sqrt{\gamma_B} + \sqrt{\gamma_S} \geq 1$, fully efficient trade is implementable for any shared value distributions. If the lying cost parameters are equal between the two, that is $\gamma_B = \gamma_S = \gamma$, these thresholds are $\gamma \geq \frac{1}{2}$ and $\gamma \geq \frac{1}{4}$, respectively.

There are many possible explanations for why someone may be averse to lying or inclined to truth-telling. Kartik 2009 discusses a lying cost in a strategic communication setting. Some motivations for introducing a lying cost in that setting, such as potential ex-post verification or an innate behavioral aversion to lying, can be transferred to this setting.

Saran 2011 considers the bilateral trade setting with the presence of "naive" agents. These agents are committed to truthful reporting, while the other "strategic" agents act strategically. In this setting, the mechanism designer can improve efficiency above the usual Myerson-Satterthwaite level by extracting rents from naive agents to compensate strategic ones. The setting I consider in this paper can be seen somewhat as a generalization, as agents in Saran 2011 have either $\gamma = 0$(strategic) or $\gamma = \infty$(naive). However, this paper will not consider γ as a type agents may realize.

It is important to note that, for my results, I restrict attention to direct bargaining mechanisms without invoking a revelation principle. This is because there is no reasonable definition of lying in an arbitrary indirect mechanism, such as an ascending auction, and thus no reasonable way to attribute a cost to lying.

The sections will be as follows. Section 2 will describe the model I use and important definitions, Sections 3, 4, and 5 will discuss incentive compatibility, individual rationality, and budget balance in that order. Section 6 will be the main results section discussing the efficient allocation. Section 7 will give concluding remarks.

2 The Model

We follow the setup in Myerson and Satterthwaite 1983, but alter the payoff definitions to include the lying cost. The convention throughout this paper will be to denote the buyer's value(unrealized, realized) by V, v and the seller's value(unrealized, realized) by C, c and the buyer's functions with a subscript B and the seller's functions with a subscript S. The buyer and seller together will be referred to generally as "agents." Denote the following:

$$F_B, \text{the buyer's value distribution} \quad F_S, \text{the seller's value distribution}$$

$$f_B, \text{the density of } F_B \quad f_S, \text{the density of } F_S$$

$$[\underline{v}, \overline{v}], \text{the support of } F_B \quad [\underline{c}, \overline{c}], \text{the support of } F_S$$

We will identify a distribution by its cumulative distribution function, so F_B and F_S are more specifically the value distributions' CDFs. The buyer and seller's values V and C are drawn independently from these distributions. f_B and f_S are assumed to be continuous and strictly positive over their supports. Their supports are also assumed to intersect at an interval of length greater than 0.

Denote the rules:

$$q : [\underline{v}, \overline{v}] \times [\underline{c}, \overline{c}] \to [0, 1], \text{ the allocation rule or probability of trade}$$

$$p_B : [\underline{v}, \overline{v}] \times [\underline{c}, \overline{c}] \to \mathbb{R}, \text{ the buyer's payment rule}$$

$$p_S : [\underline{v}, \overline{v}] \times [\underline{c}, \overline{c}] \to \mathbb{R}, \text{the seller's payment rule}$$

Note that we do not require $p_B = p_S$. Any difference at the end of the trade will go to or be taken from an intermediary. We will consider ourselves, the designer, the intermediary.

Throughout this paper, we will denote a buyer's report as v' and a seller's report as c'. Denote the interim values given the other agent reports truthfully:

$$Q_B(v') = \int_{\underline{c}}^{\overline{c}} q(v', c) f_S(c)\, dc \quad Q_S(c') = \int_{\underline{v}}^{\overline{v}} q(v, c') f_B(v)\, dv$$

$$P_B(v') = \int_{\underline{c}}^{\overline{c}} p_B(v', c) f_S(c)\, dc \quad P_S(c') = \int_{\underline{v}}^{\overline{v}} p_S(v, c') f_B(v)\, dv$$

$$U_B(v, v') = vQ_B(v') - P_B(v') - \gamma_B|v - v'| \quad U_S(c, c') = P_S(c') - cQ_S(c') - \gamma_S|c - c'|$$

where $\gamma_B, \gamma_S \geq 0$ are the buyer and seller's lying cost parameters, respectively. So, Q gives the agent's interim probability of trade given his/her report, P gives the agent's interim expected payment given his/her report, and U gives the agent's interim expected payoff given his/her true value and report. For compactness, we will write $U(t, t)$ as $U(t)$.

With these, we can define important terms to be used throughout this paper.

Definition 1. *A **direct bargaining mechanism**(or **mechanism** for short, in this paper) is a triple that consists of an allocation rule q, the buyer's payment rule p_B, and the seller's payment rule p_S.*

With the standard quasi-linear utilities with the lying costs, a direct bargaining mechanism induces a game in which each agent learns only his/her value, and then reports it to the mechanism, receiving payoffs based on the rules and the agents' values. For convenience, we will also refer to this game as the mechanism.

I reiterate that we will only consider direct bargaining mechanisms without using the revelation principle as justification because "lying" is unreasonable to define over all possible mechanisms. Since it is common to interpret actions in direct mechanisms as "reporting a type," we can reasonably define this lying cost for direct bargaining mechanisms.

Definitions 2, 3, 4, and 5 describe desirable qualities of a mechanism.

Definition 2. *A mechanism is **(Bayesian) incentive compatible(IC)** if and only if truthful reporting is a Bayesian Nash equilibrium. That is, for all $v, v' \in [\underline{v}, \overline{v}]$, $U_B(v) \geq U_B(v, v')$ and for all $c, c' \in [\underline{c}, \overline{c}]$, $U_S(c) \geq U_S(c, c')$.*

The second sentence states that when the other agent reports truthfully, the first agent finds it optimal to also report truthfully. We will call such an equilibrium a *truthful equilibrium*.

Among direct bargaining mechanisms, we can further restrict attention to IC direct bargaining mechanisms. See Kephart and Conitzer 2021 Definition 5.1 and notice that the lying cost satisfies the VTFU condition. Since truthful reporting has a lying cost of 0, ($bb = 0$, in the paper's formulation), the VTFU condition becomes for all $a, b, x \in \mathbb{R}$ $\gamma_i|a - x| \leq \gamma_i|a - b| + \gamma_i|b - x|$, for all $i \in \{B, S\}$. This is satisfied by the triangle inequality, so Theorem 5.2 justifies the restriction with a more general version of the revelation principle.

Definition 3. *An IC mechanism is **(interim) individually rational(IR)** if and only if in truthful equilibrium, agents' expected utility after learning their value is greater than or equal to 0. That is, for all $v \in [\underline{v}, \overline{v}]$, $U_B(v) \geq 0$ and for all $c \in [\underline{c}, \overline{c}]$, $U_S(c) \geq 0$.*

IR is a participation constraint requiring that each agents, after learning his/her own value, prefers to participate in the mechanism to the outside option of no trade and no payment.

Definition 4. *An IC mechanism is **(weak, ex ante) budget balanced(BB)** if and only if in truthful equilibrium, its expected revenue is greater than or equal to 0. That is, $\mathbb{E}[p_B(V, C) - p_S(V, C)] \geq 0$.*

This quality is desirable because as a designer, we would like to not have to subsidize the trade, on average. Note this definition is specifically for weak, ex-ante budget balance. Requiring this version of budget balance does not change achievable allocations and utilities from requiring weak, ex-post budget balance. The result by Börgers and Norman 2009, shows that, under independent value distributions, interim payments for every report can be preserved while correcting weak, ex-ante budget balance into weak, ex-post budget balance. Then, since interim payments are preserved for every report without changing the allocation rule or lying costs, so are interim utilities.

Definition 5. *An IC mechanism is* ***ex-post efficient****(or* ***efficient*** *for short, in this paper) if and only if in truthful equilibrium, it maximizes welfare across the buyer, seller, and designer for all realized values. That is, its allocation rule is* $q^*(v,c) = \mathbb{1}(v \geq c)$.

We will call $q^*(v,c) = \mathbb{1}(v \geq c)$ the *efficient allocation rule.*

Definition 6. *An allocation rule* q *is* ***implementable*** *if and only if there exist payment rules* p_B, p_S *such that the mechanism* (q, p_B, p_S) *is IC, IR, and BB.*

To analyze these qualities, since the buyer and seller are symmetrical in a way, I will explicitly derive conditions for the buyer, and state the analogous seller conditions.

3 Incentive Compatibility

For incentive compatibility, we start with the condition in its definition. For all $v' \in [\underline{v}, \overline{v}]$ and for all $c' \in [\underline{c}, \overline{c}]$

$$U_B(v) \geq U_B(v, v') \text{ and } U_S(c) \geq U_S(c, c').$$

This means for all $v_1, v_2 \in [\underline{v}, \overline{v}]$ with $v_1 < v_2$,

$$\begin{aligned} & U_B(v_1) \geq U_B(v_1, v_2) \\ \iff & U_B(v_1) \geq U_B(v_2) - (v_2 - v_1)Q_B(v_2) - \gamma_B|v_1 - v_2| \end{aligned}$$

and

$$\begin{aligned} & U_B(v_2) \geq U_B(v_2, v_1) \\ \iff & U_B(v_2) \geq U_B(v_1) - (v_1 - v_2)Q_B(v_1) - \gamma_B|v_2 - v_1|. \end{aligned}$$

Together, the two inequalities form the condition

$$\begin{aligned} & (v_2 - v_1)Q_B(v_1) - \gamma_B(v_2 - v_1) \leq U_B(v_2) - U_B(v_1) \leq (v_2 - v_1)Q_B(v_2) + \gamma_B(v_2 - v_1) \\ \iff & (v_2 - v_1)(Q_B(v_1) - \gamma_B) \leq U_B(v_2) - U_B(v_1) \leq (v_2 - v_1)(Q_B(v_2) + \gamma_B) \qquad (*) \\ \iff & Q_B(v_1) - \gamma_B \leq \frac{U_B(v_2) - U_B(v_1)}{v_2 - v_1} \leq Q_B(v_2) + \gamma_B \end{aligned}$$

Going through the same calculations for the seller, we get a similar condition, for all values $c_1 < c_2$:

$$-Q_S(c_1) - \gamma_S \leq \frac{U_S(c_2) - U_S(c_1)}{c_2 - c_1} \leq -Q_S(c_2) + \gamma_S$$

Thus, by taking limits with $v_1 \to v_2$, where U_B and U_S are differentiable, their derivatives must satisfy

$$Q_B(v) - \gamma_B \leq U_B'(v) \leq Q_B(v) + \gamma_B \text{ and } -Q_S(c) - \gamma_S \leq U_S'(c) \leq -Q_S(c) + \gamma_S.$$

These derivative conditions are necessary but not sufficient, generally. When we take limits to pass to the derivative, we lose our chain of equivalences. Notice that $(*)$ implies Lipschitz continuity with Lipschitz constant $1+\gamma_B$ for U_B since Q_B is at most 1 and at least 0. Importantly, this means U_B is absolutely continuous so is integrable and satisfies the fundamental theorem of calculus. Additionally, absolute continuity implies differentiability almost everywhere, so its derivative condition is satisfied almost everywhere. A similar reasoning shows this for U_S and its derivative condition.

For a sufficient condition, suppose Q_B is weakly increasing and U_B is absolutely continuous and satisfies its derivative condition almost everywhere. Integrating the derivative condition over $[v_1, v_2]$ with $v_1 < v_2$ gives

$$\int_{v_1}^{v_2} Q_B(v) - \gamma_B \, dv \leq \int_{v_1}^{v_2} U_B'(v) \, dv \leq \int_{v_1}^{v_2} Q_B(v) + \gamma_B \, dv$$

$$\implies (v_2 - v_1)(Q_B(v_1) - \gamma_B) \leq U_B(v_2) - U_B(v_1) \leq (v_2 - v_1)(Q_B(v_2) + \gamma_B)$$

since Q_B is weakly increasing. This is the same condition as $(*)$. Likewise, if Q_S is weakly decreasing and U_S is absolutely continuous and satisfies its derivative condition almost everywhere, we have the seller's version of this condition by a similar reasoning. Summarizing,

Proposition 1. *If Q_B is weakly increasing and Q_S is weakly decreasing, then a mechanism is incentive compatible if and only if U_B and U_S are absolutely continuous and for almost every $v \in [\underline{v}, \overline{v}]$ and for almost every $c \in [\underline{c}, \overline{c}]$,*

$$Q_B(v) - \gamma_B \leq U_B'(v) \leq Q_B(v) + \gamma_B \text{ and } -Q_S(c) - \gamma_S \leq U_S'(c) \leq -Q_S(c) + \gamma_S. \quad (1)$$

4 Individual Rationality

Now, suppose we have an incentive compatible mechanism with Q_B weakly increasing and Q_S weakly decreasing. Individual rationality requires for all $v \in [\underline{v}, \overline{v}]$ and for all $c \in [\underline{c}, \overline{c}]$,

$$U_B(v) \geq 0 \text{ and } U_S(c) \geq 0.$$

Since our mechanism is incentive compatible, (1) implies for almost every $v \in [\underline{v}, \overline{v}]$,

$$U_B'(v) \geq Q_B(v) - \gamma_B.$$

Integrating over $[v_1, v_2]$ with $v_1 < v_2$,

$$\int_{v_1}^{v_2} U_B'(v) \, dv \geq \int_{v_1}^{v_2} Q_B(v) - \gamma_B \, dv$$

$$\implies U_B(v_2) - U_B(v_1) \geq \int_{v_1}^{v_2} Q_B(v) - \gamma_B \, dv$$

$$\implies U_B(v_2) \geq \int_{v_1}^{v_2} Q_B(v) - \gamma_B \, dv,$$

since $U_B(v_1) \geq 0$. Since our selection of $v_1 < v_2$ was arbitrary,

$$U_B(v_2) \geq \sup_{v_1 \in [\underline{v}, v_2]} \int_{v_1}^{v_2} Q_B(v) - \gamma_B \, dv. \tag{$**$}$$

If on $[\underline{v}, v_2]$ $Q_B(v) - \gamma_B$ is sometimes positive and sometimes negative our optimal choice v_1^* is a point such that before it, $Q_B(v) - \gamma_B \leq 0$ and after it $Q_B(v) - \gamma_B \geq 0$ which exists since Q_B is weakly increasing. If $Q_B(v) - \gamma_B$ is always positive or 0, $v_1^* = \underline{v}$ and if $Q_B(v) - \gamma_B$ is always negative, $v_1^* = v_2$. Thus,

$$\sup_{v_1 \in [\underline{v}, v_2]} \int_{v_1}^{v_2} Q_B(v) - \gamma_B \, dv = \int_{\underline{v}}^{v_2} \max \{Q_B(v) - \gamma_B, 0\} \, dv.$$

Again, since our selection of $v_1 < v_2$ was arbitrary, $(**)$ and the above give for all $v \in [\underline{v}, \overline{v}]$,

$$U_B(v) \geq \int_{\underline{v}}^{v} \max \{Q_B(t) - \gamma_B, 0\} \, dt$$

A similar reasoning for the seller gives for all $c \in [\underline{c}, \overline{c}]$

$$U_S(c) \geq \int_{c}^{\overline{c}} \max \{Q_S(t) - \gamma_S, 0\} \, dt$$

A mechanism that attains equality is also incentive compatible and individually rational. For individual rationality, the integrand is nonnegative, and the limits are in increasing order. For incentive compatibility, notice that when equality is attained, these are integrals of bounded functions and therefore absolutely continuous, and for almost every $v \in [\underline{v}, \overline{v}]$ and for almost every $c \in [\underline{c}, \overline{c}]$,

$$U_B'(v) = \max \{Q_B(v) - \gamma_B, 0\} \text{ and } U_S'(c) = -\max \{Q_S(c) - \gamma_S, 0\}.$$

These derivatives satisfy (1) by construction. Thus, we have the following proposition.

Proposition 2. *An incentive compatible mechanism with Q_B weakly increasing and Q_S weakly decreasing is individually rational if and only if for all $v \in [\underline{v}, \overline{v}]$ and for all $c \in [\underline{c}, \overline{c}]$*

$$U_B(v) \geq \int_{\underline{v}}^{v} \max \{Q_B(t) - \gamma_B, 0\} \, dt \text{ and } U_S(c) \geq \int_{c}^{\overline{c}} \max \{Q_S(t) - \gamma_S, 0\} \, dt.$$

In particular, a mechanism with Q_B weakly increasing and Q_S weakly decreasing and

$$U_B(v) = \underline{U}_B(v) \coloneqq \int_{\underline{v}}^{v} \max \{Q_B(t) - \gamma_B, 0\} \, dt,$$

$$U_S(c) = \underline{U}_S(c) \coloneqq \int_{c}^{\overline{c}} \max \{Q_S(t) - \gamma_S, 0\} \, dt.$$

is incentive compatible and individually rational.

5 Budget Balance

Given an allocation rule and interim utilities, we can express expected revenue under truthful reporting without explicitly specifying payments first. Recall that, by definition, $P_B(v) = vQ_B(v) - U_B(v)$ and $P_S(c) = cQ_S(c) + U_S(c)$, so

$$\begin{aligned}
\mathbb{E}[p_B(V,C)] - \mathbb{E}[p_S(V,C)] &= \mathbb{E}[P_B(V)] - \mathbb{E}[P_S(C)] \\
&= \mathbb{E}[VQ_B(V) - U_B(V)] - \mathbb{E}[CQ_S(C) + U_S(C)] \\
&= \mathbb{E}[VQ_B(V)] - \mathbb{E}[U_B(V)] - \mathbb{E}[CQ_S(C)] - \mathbb{E}[U_S(C)] \\
&= \mathbb{E}[Vq(V,C)] - \mathbb{E}[U_B(V)] - \mathbb{E}[Cq(V,C)] - \mathbb{E}[U_S(C)] \\
&= \mathbb{E}[(V-C)q(V,C)] - (\mathbb{E}[U_B(V)] + \mathbb{E}[U_S(C)]).
\end{aligned}$$

In the final expression, the first term is expected gains from trade. The second is the agents' expected utility. Given an allocation rule, expected gains from trade is fixed, but we have freedom over how much utility we give to the agents. To increase expected revenue, we should choose lower interim utilities to give to the agents. By Proposition 2, for an allocation rule with Q_B weakly increasing and Q_S weakly decreasing, the lowest we can choose while preserving IC, IR are $\underline{U}_B$ and $\underline{U}_S$. Then, for such an allocation rule, it suffices to check the budget balance condition for these utilities, since it maximizes expected revenue under IC, IR.

Given an allocation rule with Q_B weakly increasing and Q_S weakly decreasing these interim utilities are also achievable by setting $p_B(v,c) = vq(v,c) - \underline{U}_B(v)$ and $p_S(v,c) = cq(v,c) + \underline{U}_S(c)$. These payments set interim payments, and thus expected revenue, as above. Furthermore, by Proposition 2, this mechanism is IC, IR since it achieves the interim utilities $\underline{U}_B$ and $\underline{U}_S$.

Then, we have that an allocation rule with Q_B weakly increasing and Q_S weakly decreasing is implementable if and only if

$$\mathbb{E}[(V-C)q(V,C)] - (\mathbb{E}[\underline{U}_B(V)] + \mathbb{E}[\underline{U}_S(C)]) \geq 0.$$

For the buyer,

$$\begin{aligned}
\mathbb{E}[\underline{U}_B(V)] &= \int_{\underline{v}}^{\overline{v}} \left(\int_{\underline{v}}^{v} \max\{Q_B(t) - \gamma_B, 0\}\, dt \right) f_B(v)\, dv \\
&= \int_{\underline{v}}^{\overline{v}} \int_{\underline{v}}^{v} \max\{Q_B(t) - \gamma_B, 0\} f_B(v)\, dt\, dv \\
&= \int_{\underline{v}}^{\overline{v}} \int_{t}^{\overline{v}} \max\{Q_B(t) - \gamma_B, 0\} f_B(v)\, dv\, dt, \text{ by Fubini's theorem} \\
&= \int_{\underline{v}}^{\overline{v}} \max\{Q_B(t) - \gamma_B, 0\} \int_{t}^{\overline{v}} f_B(v)\, dv\, dt \\
&= \int_{\underline{v}}^{\overline{v}} \max\{Q_B(t) - \gamma_B, 0\}(1 - F_B(t))\, dt
\end{aligned}$$

$$= \int_{\underline{v}}^{\overline{v}} \max\{Q_B(v) - \gamma_B, 0\}(1 - F_B(v))\, dv$$

For the use of Fubini's theorem, notice f_B is assumed to be continuous on a compact interval so is bounded, and $\max\{Q_B(t) - \gamma_B, 0\}$ is also bounded. Then, the integrand is absolutely integrable so switching integration order is justified.

A similar calculation for the seller gives

$$\mathbb{E}[\underline{U}_S(C)] = \int_{\underline{c}}^{\overline{c}} \max\{Q_S(c) - \gamma_S, 0\} F_S(c)\, dc.$$

And thus we have,

Proposition 3. *An allocation rule with Q_B weakly increasing and Q_S weakly decreasing is implementable if and only if*

$$\mathbb{E}[(V - C)q(V, C)] - (\mathbb{E}[\underline{U}_B(V)] + \mathbb{E}[\underline{U}_S(C)]) \geq 0.$$

Furthermore,

$$\mathbb{E}[\underline{U}_B(V)] = \int_{\underline{v}}^{\overline{v}} \max\{Q_B(v) - \gamma_B, 0\}(1 - F_B(v))\, dv,$$

$$\mathbb{E}[\underline{U}_S(C)] = \int_{\underline{c}}^{\overline{c}} \max\{Q_S(c) - \gamma_S, 0\} F_S(c)\, dc.$$

6 The Efficient Allocation

The efficient allocation rule $q^*(v, c) = \mathbb{1}(v \geq c)$ has interim allocation probabilities $Q_B(v) = F_S(v)$ and $Q_S(c) = 1 - F_B(c)$. We can calculate the efficient allocation rule's expected gains from trade G^* as a function of F_B and F_S.

$$\begin{aligned}
G^*(F_B, F_S) &= \mathbb{E}[(V - C)\mathbb{1}(V \geq C)] \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (v - c)\mathbb{1}(v \geq c) f_S(c) f_B(v)\, dc\, dv \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \mathbb{1}(c \leq t)\mathbb{1}(t \leq v)\, dt \right) f_S(c) f_B(v)\, dc\, dv \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \mathbb{1}(c \leq t) f_S(c)\, dc \right) \left(\int_{-\infty}^{\infty} \mathbb{1}(t \leq v) f_B(v)\, dv \right) dt \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{t} f_S(c)\, dc \right) \left(\int_{t}^{\infty} f_B(v)\, dv \right) dt \\
&= \int_{-\infty}^{\infty} F_S(t)(1 - F_B(t))\, dt.
\end{aligned}$$

For the third equality, recall that the layer-cake representation of $v - c$ when $v \geq c$ is

$$v - c = \int_c^v dt = \int_{-\infty}^{\infty} \mathbb{1}(c \leq t \leq v)\, dt = \int_{-\infty}^{\infty} \mathbb{1}(c \leq t)\mathbb{1}(t \leq v)\, dt.$$

And the fourth equality is justified by Fubini's theorem again.

By Proposition 3, the agents' minimum expected utilities (while remaining IC, IR) U_B^*, U_S^* as functions of γ_B, F_B, F_S and γ_S, F_B, F_S are

$$U_B^*(\gamma_B, F_B, F_S) = \int_{\underline{v}}^{\overline{v}} \max\{F_S(v) - \gamma_B, 0\}(1 - F_B(v))\, dv,$$

$$U_S^*(\gamma_S, F_B, F_S) = \int_{\underline{c}}^{\overline{c}} \max\{1 - F_B(c) - \gamma_S, 0\}F_S(c)\, dc.$$

Since $F_S(v)$ is weakly increasing and $1 - F_B(c)$ weakly decreasing by definition, we can use Proposition 3. Then, the efficient allocation rule is implementable if and only if

$$G^*(F_B, F_S) - (U_B^*(\gamma_B, F_B, F_S) + U_S^*(\gamma_S, F_B, F_S)) \geq 0 \tag{2}$$

Note for the following results that U_B^* and U_S^* are weakly decreasing with γ_B and γ_S, respectively, so the LHS of (2) is weakly increasing with γ_B and γ_S. Thus, for all $\gamma_{B,1} \leq \gamma_{B,2}, \gamma_{S,1} \leq \gamma_{S,2}$ if (2) is satisfied for $\gamma_{B,1}, \gamma_{S,1}$, then it is also satisfied for all $\gamma_{B,2}, \gamma_{S,2}$

This first result's reverse direction restates the Myerson-Satterthwaite Theorem.

Proposition 4 (The Myerson-Satterthwaite Theorem)**.** *The efficient allocation rule is not implementable for any valid distributions if and only if* $\gamma_B = \gamma_S = 0$.

Proof. Start with the reverse direction. For some distributions F_B and F_S, agents' minimum total expected utility when $\gamma_B = \gamma_S = 0$ is

$$\begin{aligned}
&U_B^*(0, F_B, F_S) + U_S^*(0, F_B, F_S) \\
&= \int_{\underline{v}}^{\overline{v}} \max\{F_S(v) - 0, 0\}(1 - F_B(v))\, dv + \int_{\underline{c}}^{\overline{c}} \max\{1 - F_B(c) - 0, 0\}F_S(c)\, dc \\
&= \int_{\underline{v}}^{\overline{v}} F_S(v)(1 - F_B(v))\, dv + \int_{\underline{c}}^{\overline{c}} (1 - F_B(c))F_S(c)\, dc,
\end{aligned}$$

and expected gains from trade are

$$G^*(F_B, F_S) = \int_{-\infty}^{\infty} F_S(t)(1 - F_B(t))\, dt.$$

So (2) is,

$$\int_{-\infty}^{\infty} F_S(t)(1 - F_B(t))\, dt - \left(\int_{\underline{v}}^{\overline{v}} F_S(v)(1 - F_B(v))\, dv + \int_{\underline{c}}^{\overline{c}} (1 - F_B(c))F_S(c)\, dc \right) \geq 0.$$

We can rewrite the LHS as

$$\int_{-\infty}^{\infty} F_S(t)(1-F_B(t))\,dt - \left(\int_{\underline{v}}^{\overline{v}} F_S(t)(1-F_B(t))\,dt + \int_{\underline{c}}^{\overline{c}} (1-F_B(t))F_S(t)\,dt\right)$$

$$= \int_{[\underline{v},\overline{v}]\cup[\underline{c},\overline{c}]} F_S(t)(1-F_B(t))\,dt - \left(\int_{[\underline{v},\overline{v}]} F_S(t)(1-F_B(t))\,dt + \int_{[\underline{c},\overline{c}]} (1-F_B(t))F_S(t)\,dt\right)$$

$$= -\int_{[\underline{v},\overline{v}]\cap[\underline{c},\overline{c}]} F_S(t)(1-F_B(t))\,dt, \text{ by the inclusion-exclusion principle.}$$

This is strictly negative by our assumptions that f_B, f_S are strictly positive on their respective supports, and that these supports intersect at a positive-length interval. Thus, the efficient allocation is not implementable.

For the forward direction, by symmetry between the buyer and seller and the monotonicity of the LHS of (2), it suffices to check implementability for $\gamma_B > 0$ and $\gamma_S = 0$. Our goal is to find distributions F_B, F_S such that we have (2). Consider when the buyer's value distribution is Uniform$[0,1]$ and the seller's is Uniform$[1-\epsilon, 2-\epsilon]$ for some $0 < \epsilon \leq 1$. These distributions have strictly positive, continuous densities on their supports, which intersect at an interval of length $\epsilon > 0$, so these are valid distributions.

To compute gains from trade, note the CDFs of these uniform distributions are

$$F_B(t) = \begin{cases} 0 & \text{if } t < 0 \\ t & \text{if } 0 \leq t \leq 1 \\ 1 & \text{if } t > 1 \end{cases} \text{ and } F_S(t) = \begin{cases} 0 & \text{if } t < 1-\epsilon \\ t-(1-\epsilon) & \text{if } 1-\epsilon \leq t \leq 2-\epsilon \\ 1 & \text{if } t > 2-\epsilon \end{cases}.$$

$$\begin{aligned} G^*_\epsilon(F_B, F_S) &= \int_{1-\epsilon}^{1} (t-(1-\epsilon))(1-t)\,dt \\ &= \int_0^{\epsilon} x(\epsilon - x)\,dx, \ x = t-(1-\epsilon) \\ &= \frac{\epsilon x^2}{2} - \frac{x^3}{3}\bigg|_{x=0}^{\epsilon} \\ &= \frac{\epsilon^3}{6}. \end{aligned}$$

Agents' minimum expected utility when $\gamma_B \leq \epsilon$ is

$$U^*_{B,\epsilon}(\gamma_B, F_B, F_S) + U^*_{S,\epsilon}(\gamma_S, F_B, F_S)$$

$$= \int_0^1 \max\{F_S(v) - \gamma_B, 0\}(1-F_B(v))\,dv + \int_{1-\epsilon}^{2-\epsilon} \max\{1-F_B(c)-\gamma_S, 0\}F_S(c)\,dc$$

$$= \int_{1-\epsilon}^1 \max\{(v-(1-\epsilon)-\gamma_B)(1-v), 0\}\,dv + \int_{1-\epsilon}^1 \max\{(1-c-\gamma_S)(c-(1-\epsilon)), 0\}\,dc$$

$$
\begin{aligned}
&= \int_{1-\epsilon+\gamma_B}^{1} (1-v)(v-(1-\epsilon)-\gamma_B)\,dv + \int_{1-\epsilon}^{1-\gamma_S} (1-c-\gamma_S)(c-(1-\epsilon))\,dc \\
&= \int_{\gamma_B}^{\epsilon} (\epsilon-x)(x-\gamma_B)\,dx + \int_{\gamma_S}^{\epsilon} (y-\gamma_S)(\epsilon-y)\,dy,\ \ x = v-(1-\epsilon), y = 1-c \\
&= \left(\frac{\epsilon x^2}{2} - \frac{x^3}{3} - \gamma_B \epsilon x + \frac{\gamma_B x^2}{2}\right)\Bigg|_{x=\gamma_B}^{\epsilon} + \left(\frac{\epsilon y^2}{2} - \frac{y^3}{3} - \gamma_S \epsilon y + \frac{\gamma_S y^2}{2}\right)\Bigg|_{y=\gamma_S}^{\epsilon} \\
&= \frac{(\epsilon-\gamma_B)^3}{6} + \frac{(\epsilon-\gamma_S)^3}{6} \\
&= \frac{(\epsilon-\gamma_B)^3}{6} + \frac{\epsilon^3}{6}.
\end{aligned}
$$

If $\gamma_B > \epsilon$, $U_B^*(\gamma_B, F_B, F_S) = 0$. Then, choose $\epsilon \leq \gamma_B$ so that $U_B^*(\gamma_B, F_B, F_S) = 0$, making $G^*(F_B, F_S) - (U_B^*(\gamma_B, F_B, F_S) + U_S^*(\gamma_S, F_B, F_S)) = 0$, satisfying (2). □

Next, we establish a threshold for the total lying cost.

Proposition 5. *If $\gamma_B + \gamma_S \geq 1$, for all F_B, F_S, for any $k \in [1-\gamma_S, \gamma_B] \cap [0,1]$, the k-double auction is incentive compatible, individually rational, budget balanced, and efficient.*

Proof. Suppose $\gamma_B + \gamma_S \geq 1$ and let $k \in [1-\gamma_S, \gamma_B] \cap [0,1]$. If the mechanism is IC, any k-double auction is clearly IR, BB, and efficient. It remains to verify that this mechanism is indeed IC.

Recall the k-double auction's payments rules are

$$p_B(v,c) = p_S(v,c) = (kv + (1-k)c)\mathbb{1}(v \geq c).$$

Fix the seller's report c', and consider the buyer's decision, denoting his payoff based on his value, his report, and the seller's report $u_B(v, v', c')$. When the buyer has value v, the difference in payoff between reporting v and reporting v' is

$$
\begin{aligned}
&u_B(v,v,c') - u_B(v,v',c') \\
&= \mathbb{1}(v \geq c')(v - (kv + (1-k)c')) - (\mathbb{1}(v' \geq c')(v - (kv' + (1-k)c')) - \gamma_B|v-v'|) \\
&= \mathbb{1}(v \geq c')((1-k)v - (1-k)c') - (\mathbb{1}(v' \geq c')(v - (k(v+v'-v) + (1-k)c')) - \gamma_B|v-v'|) \\
&= (1-k)(v-c')(\mathbb{1}(v \geq c') - \mathbb{1}(v' \geq c')) - (\mathbb{1}(v' \geq c')(k(v-v')) - \gamma_B|v-v'|).
\end{aligned}
$$

The second term is always weakly negative because $k \leq \gamma_B$, so subtracting it contributes weakly positively. For the first term, notice that when the differences $(\mathbb{1}(v \geq c') - \mathbb{1}(v' \geq c'))$ and $(v - c')$ are both nonzero, they have the same sign as each other, so the first term is always weakly positive. A similar reasoning shows the same for the seller, noting that $1 - k \leq \gamma_S$. Thus, it is a weakly dominant strategy to report truthfully in this mechanism for both the buyer and seller, so the mechanism is incentive compatible. □

So, when we have a sufficiently high total lying cost, we can safely use the k-double auction mechanism analyzed in Chatterjee and Samuelson 1983, adjusting k so that each agent's lying cost sufficiently offsets the potential benefit from misreporting his/her value.

In fact, $\gamma_B + \gamma_S = 1$ is the greatest lower-bound for the efficient allocation rule to be implementable across all distributions.

Corollary 1. *The efficient allocation rule is implementable for all valid distributions if and only if* $\gamma_B + \gamma_S \geq 1$.

Proof. The reverse direction follows directly from Proposition 5.

For the forward direction, let $\gamma_B + \gamma_S < 1$. Consider distributions with the following densities $f_{B,n}(v) = h_n(1-v)$ and $f_{S,n}(c) = h_n(c)$ where

$$h_n(t) = \frac{ne^{-nt}}{1-e^{-n}},\ t \in [0,1].$$

These densities are valid: they both integrate to 1, and they are both strictly positive and continuous on the same support $[0,1]$. For each distribution, as n increases, probability mass becomes increasingly concentrated at an endpoint of the support.

For every $t \in (0,1)$, as $n \to \infty$, $F_{B,n}(t) \to 0$ and $F_{S,n}(t) \to 1$. Notice that the integrands of G_n^*, $U_{B,n}^*$, and $U_{S,n}^*$ are all bounded by 1 for all n, and that this bound is clearly integrable. The Dominated Convergence Theorem allows us to exchange limits with integration, so we get that gains from trade $G_n^* \to 1$, the buyer's minimum expected utility $U_{B,n}^* \to 1-\gamma_B$, and the seller's minimum expected utility $U_{S,n}^* \to 1-\gamma_S$. Then, as $n \to \infty$ the LHS of (2)

$$G_n^* - (U_{B,n}^* + U_{S,n}^*) \to 1 - (2 - (\gamma_B + \gamma_S)) < 0,$$

since $0 \leq \gamma_B + \gamma_S < 1$. Then, choose n large enough so that $G_n^* - (U_{B,n}^* + U_{S,n}^*) < 0$, failing (2). □

Proposition 6. *The efficient allocation rule is implementable for all valid distributions with* $F_B = F_S$ *if and only if* $\sqrt{\gamma_B} + \sqrt{\gamma_S} \geq 1$.

Proof. Suppose $F_B = F_S = F$ with support $[\underline{t}, \bar{t}]$. Start with the reverse direction. Suppose $\sqrt{\gamma_B} + \sqrt{\gamma_S} \geq 1$. The LHS of (2) is

$$G^*(F,F) - (U_B^*(\gamma_B, F, F) + U_S^*(\gamma_S, F, F))$$

$$= \int_{\underline{t}}^{\bar{t}} F(t)(1-F(t)) - (\max\{F(t)-\gamma_B, 0\}(1-F(t)) + \max\{1-F(t)-\gamma_S, 0\}F(t))\, dt.$$

Denote each term in the integrand

$$G(t) = F(t)(1-F(t)),$$

$$U(t) = \max\{F(t)-\gamma_B, 0\}(1-F(t)) + \max\{1-F(t)-\gamma_S, 0\}F(t).$$

At any $t \in [\underline{t}, \bar{t}]$, if either $\max\{\}$ evaluates to 0, $U(t) \leq F(t)(1-F(t)) = G(t)$. If, instead, both evaluate to a positive expression,

$$\begin{aligned} U(t) &= (F(t) - \gamma_B)(1 - F(t)) + (1 - F(t) - \gamma_S)F(t) \\ &= 2F(t)(1 - F(t)) - \gamma_B(1 - F(t)) - \gamma_S F(t). \end{aligned}$$

To show $U(t) \leq G(t)$, it remains to show that $F(t)(1-F(t)) \leq \gamma_B(1-F(t)) + \gamma_S F(t)$. Define the vectors $\mathbf{a} = \langle \sqrt{F(t)\gamma_S}, \sqrt{(1-F(t))\gamma_B} \rangle$ and $\mathbf{b} = \langle \sqrt{1-F(t)}, \sqrt{F(t)} \rangle$. By the Cauchy-Schwarz inequality,

$$\begin{aligned} & (\mathbf{a} \cdot \mathbf{b})^2 \leq \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \\ \iff & F(t)(1-F(t))(\sqrt{\gamma_S} + \sqrt{\gamma_B})^2 \leq F(t)\gamma_S + (1-F(t))\gamma_B \\ \implies & F(t)(1-F(t)) \leq F(t)\gamma_S + (1-F(t))\gamma_B, \text{ since } \sqrt{\gamma_S} + \sqrt{\gamma_B} \geq 1. \end{aligned}$$

Thus, for all $t \in [\underline{t}, \bar{t}]$, $U(t) \leq G(t)$, so

$$G^*(F,F) - (U_B^*(\gamma_B, F, F) + U_S^*(\gamma_S, F, F)) = \int_{\underline{t}}^{\bar{t}} G(t) - U(t)\, dt \geq 0.$$

Then, (2) is satisfied.

For the forward direction, let $\sqrt{\gamma_B} + \sqrt{\gamma_S} < 1$. Define $p = \frac{1+\gamma_B-\gamma_S}{2}$. Also, define h_n as in the proof of Corollary 1. Consider the distribution with density $f_{B,n}(t) = f_{S,n}(t) = ph_n(t) + (1-p)h_n(1-t)$. Notice that $p - \gamma_B = 1 - \gamma_S - p = \frac{1-\gamma_B-\gamma_S}{2} > 0$, since $\sqrt{\gamma_B} + \sqrt{\gamma_S} < 1 \implies \gamma_B + \gamma_S < 1$. Then, $p \in (\gamma_B, 1-\gamma_S) \subseteq (0,1)$, so this density is strictly positive on its support $[0,1]$ with p of the probability mass becoming increasingly concentrated at 0. Thus, for any $t \in (0,1)$, as $n \to \infty$, $F_n(t) \to p$. Using a similar dominated convergence argument as in the proof of Corollary 1, we have that

$$G_n^* - (U_{B,n}^* + U_{S,n}^*) \to p(1-p) - ((p-\gamma_B)(1-p) + (1-p-\gamma_S)p)$$

since $\gamma_B < p < 1 - \gamma_S$, so both $\max\{\}$'s in $U(t)$ evaluate to a positive expression at $F(t) = p$. Calculating the limit expression,

$$\begin{aligned} p(1-p) - ((p-\gamma_B)(1-p) + (1-p-\gamma_S)p) &= -p(1-p) + \gamma_B(1-p) + \gamma_S p \\ &= p^2 + (-1 - \gamma_B + \gamma_S)p + \gamma_B \\ &= p^2 - 2p^2 + \gamma_B \\ &= -\frac{(1+\gamma_B-\gamma_S)^2}{4} + \gamma_B \\ &= -\frac{(1-\gamma_B-\gamma_S)^2 + 4\gamma_B(1-\gamma_S)}{4} + \gamma_B \\ &= -\frac{(1-\gamma_B-\gamma_S)^2 - 4\gamma_B\gamma_S}{4} \\ &< 0, \end{aligned}$$

since $\sqrt{\gamma_B} + \sqrt{\gamma_S} < 1 \implies \gamma_B + 2\sqrt{\gamma_B \gamma_S} + \gamma_S < 1 \implies 4\gamma_B\gamma_S < (1 - \gamma_B - \gamma_S)^2$.

Then, $G_n^* - (U_{B,n}^* + U_{S,n}^*)$ converges to something negative. Then, choose n large enough so that $G_n^* - (U_{B,n}^* + U_{S,n}^*) < 0$, failing (2). □

7 Concluding Remarks

With a sufficiently high, commonly-known cost of lying between the buyer and seller, the efficient allocation rule can be implemented. The key freedom granted from the lying cost is that rents from up to γ_B of interim allocation probability can be fully extracted from the buyer and likewise up to γ_S from the seller.

Within this setting, we may want to compute second-best allocations, given the first-best efficient allocation is not implementable. A natural extension of this setting would be to include γ as a type that would also need to be screened. It remains to determine how this would be handled. Are more lying-averse types also averse to lying about their lying aversion? Saran 2011 handles this in their two-type setting by forcing naive types to also report that they are naive naively. Strategic types are allowed to report that they are naive, so the mechanism must incentivize them to report that they are strategic instead of naive.

Without introducing lying cost as a type, it is unlikely this setting is suitable for real-life behavioral interpretation. Mainly, it is unlikely that any bilateral trade setting in reality has both parties knowing the exact lying cost of the other party, knowing that the other knows exactly theirs, and so on. Nevertheless, multidimensional types are considerably more difficult to work with, even if there is a reasonable way to handle the issue in the previous paragraph, so it is beyond the scope of this analysis.

It is worth noting that Proposition 5 which shows, under suitable conditions, dominant-strategy incentive compatibility of the k-double auction seen originally in Chatterjee and Samuelson 1983 is robust to removing the common knowledge assumption of the lying cost parameters, as long as an appropriate k is able to be chosen by the designer.

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